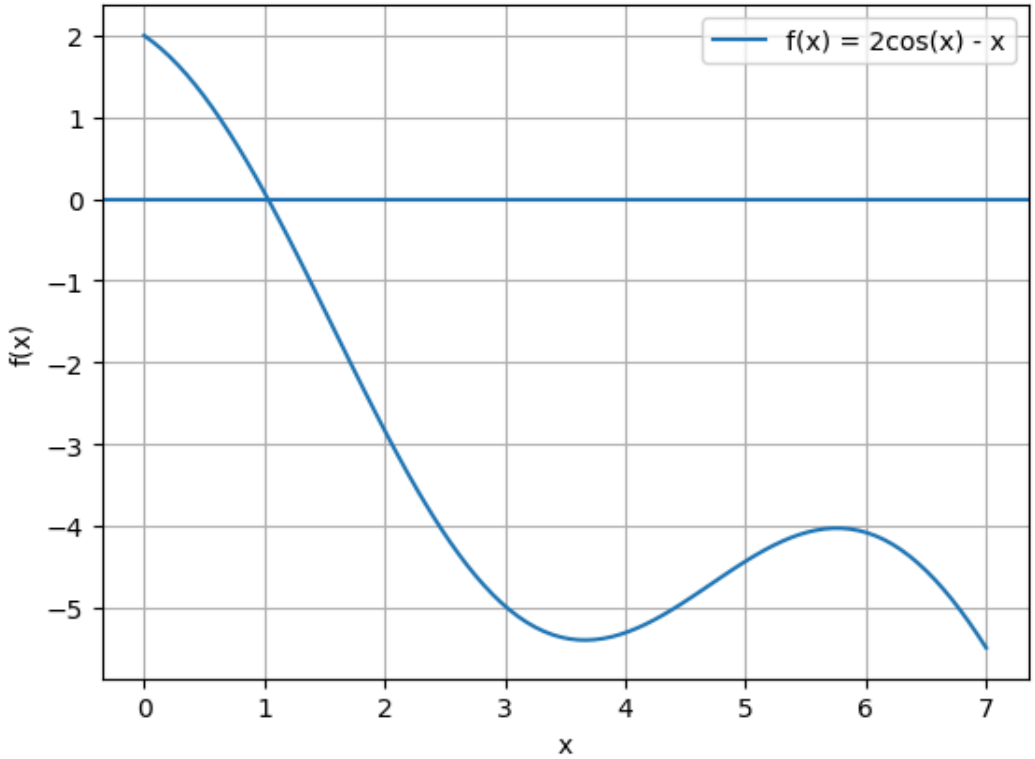
Bisection Method

Sanjeev K
ph25mscst11026

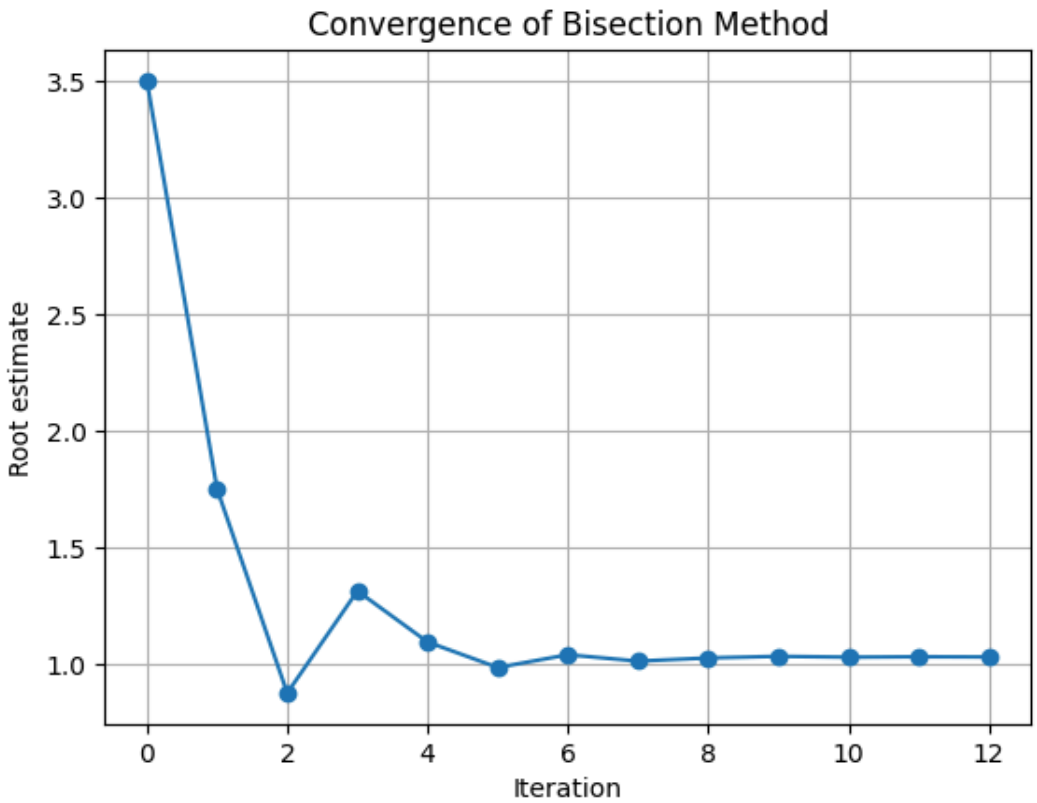
#Solution 1.

```
import numpy as np
import matplotlib.pyplot as plt

#Define Function
def f(x):
  return 2 * np.cos(x) - x

#Take values of x in the range[0, 7]
x = np.linspace(0, 7, 100)
y = f(x)

#plot the function
plt.plot(x, y)
plt.axhline()
plt.xlabel('x')
plt.ylabel('f(x)')
plt.grid(True)
plt.legend(['f(x) = 2cos(x) - x'])
plt.show()

#finding the roots of f(x) = 0 using bisection method
eps = 1e-3
Nmax = 100  #Maximum number of iterations
a = 0.0
b = 7.0

#Define Bisection Method
def Bisection(Xminus, Xplus, Nmax, eps):
    roots = []

    for it in range(Nmax):
        x = (Xplus + Xminus) / 2.0
```

```
        roots.append(x)

        if f(Xplus) * f(x) > 0.0:
            Xplus = x
        else:
            Xminus = x

        if abs(f(x)) < eps:
            print("Converged in", it+1, "iterations")
            break

    return np.array(roots)

roots = Bisection(a, b, Nmax, eps)

plt.figure()
plt.plot(range(len(roots)), roots, marker='o')
plt.xlabel("Iteration")
plt.ylabel("Root estimate")
plt.title("Convergence of Bisection Method")
plt.grid(True)
plt.show()

print(roots)
print(f'root of the f(x) is {roots[-1]}')   #1.0296630859375
print(f'value of f(x) at the root: {f(roots[-1])}')
#0.0005522177242860415
print(f'is value of the fn at the root less then eps: {f(roots[-1]) <
eps}') #True
```



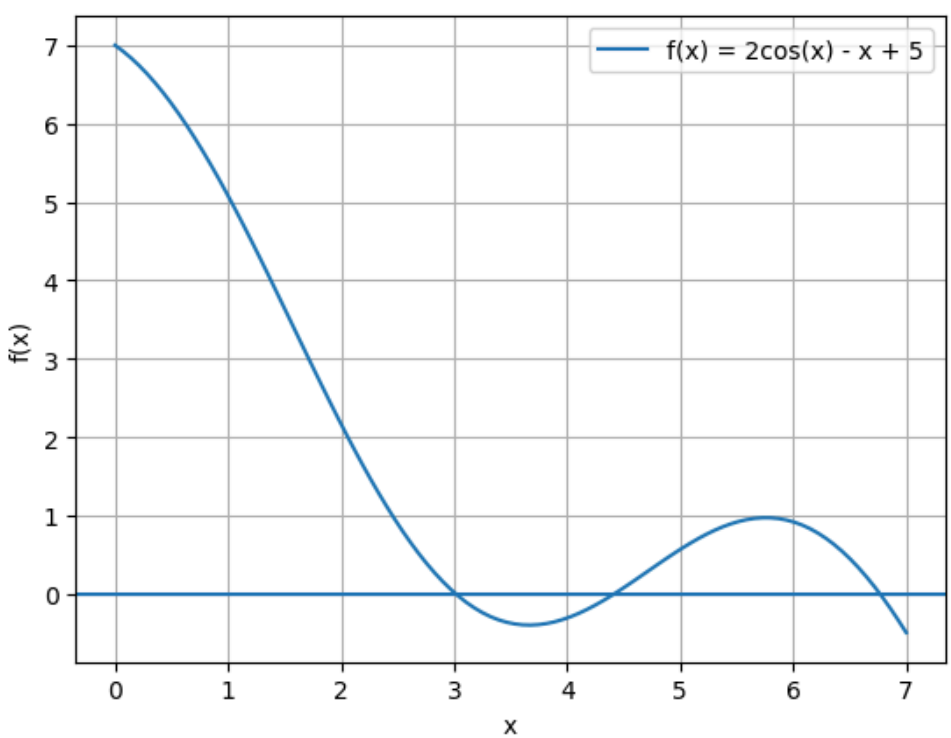

```
Converged in 13 iterations
```




```
[3.5        1.75       0.875      1.3125     1.09375    0.984375
 1.0390625  1.01171875 1.02539062 1.03222656 1.02880859 1.03051758
 1.02966309]
root of the f(x) is 1.0296630859375
value of f(x) at the root: 0.0005522177242860415
is value of the fn at the root less then eps: True
```

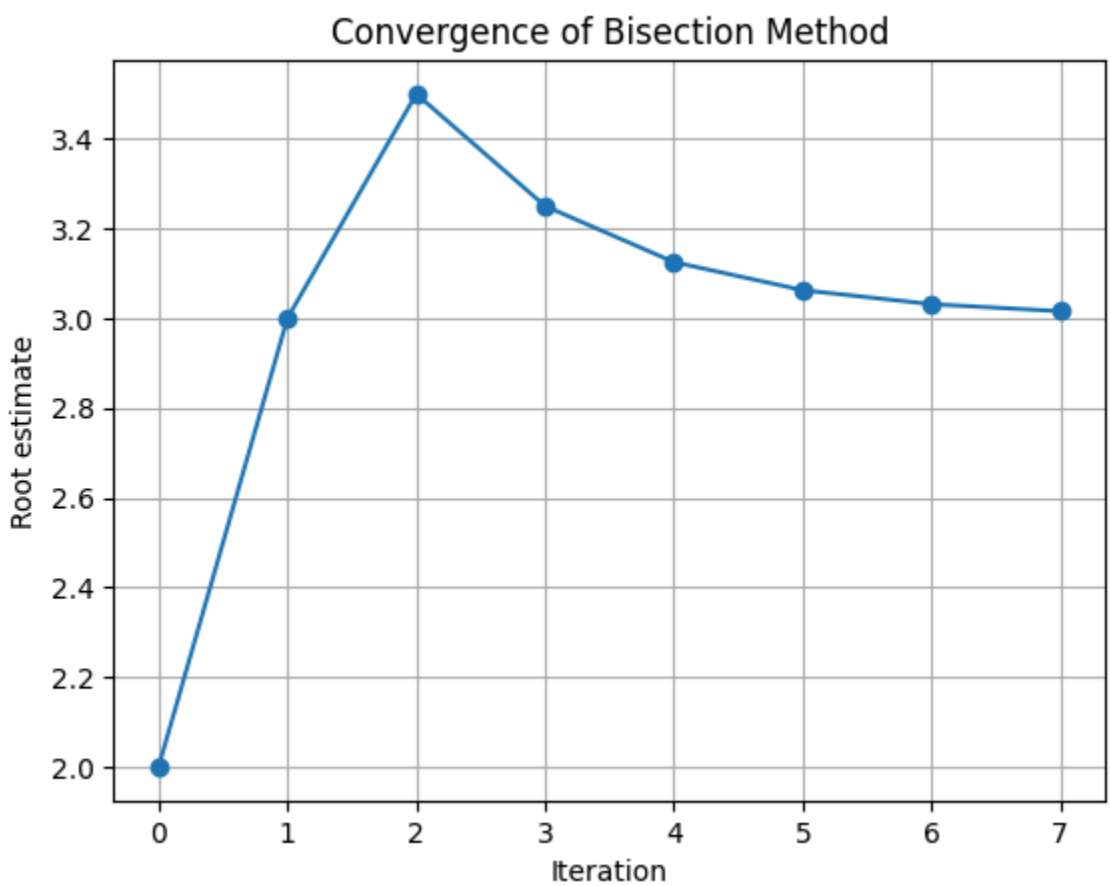
#Solution 2.

```
import numpy as np
import matplotlib.pyplot as plt

#Define Function
def f(x):
  return 2 * np.cos(x) - x + 5

#Take values of x in the range[0, 7]
x = np.linspace(0, 7, 100)
y = f(x)

#plot the function
plt.plot(x, y)
plt.axhline()
plt.xlabel('x')
plt.ylabel('f(x)')
plt.grid(True)
plt.legend(['f(x) = 2cos(x) - x + 5'])
plt.show()
```

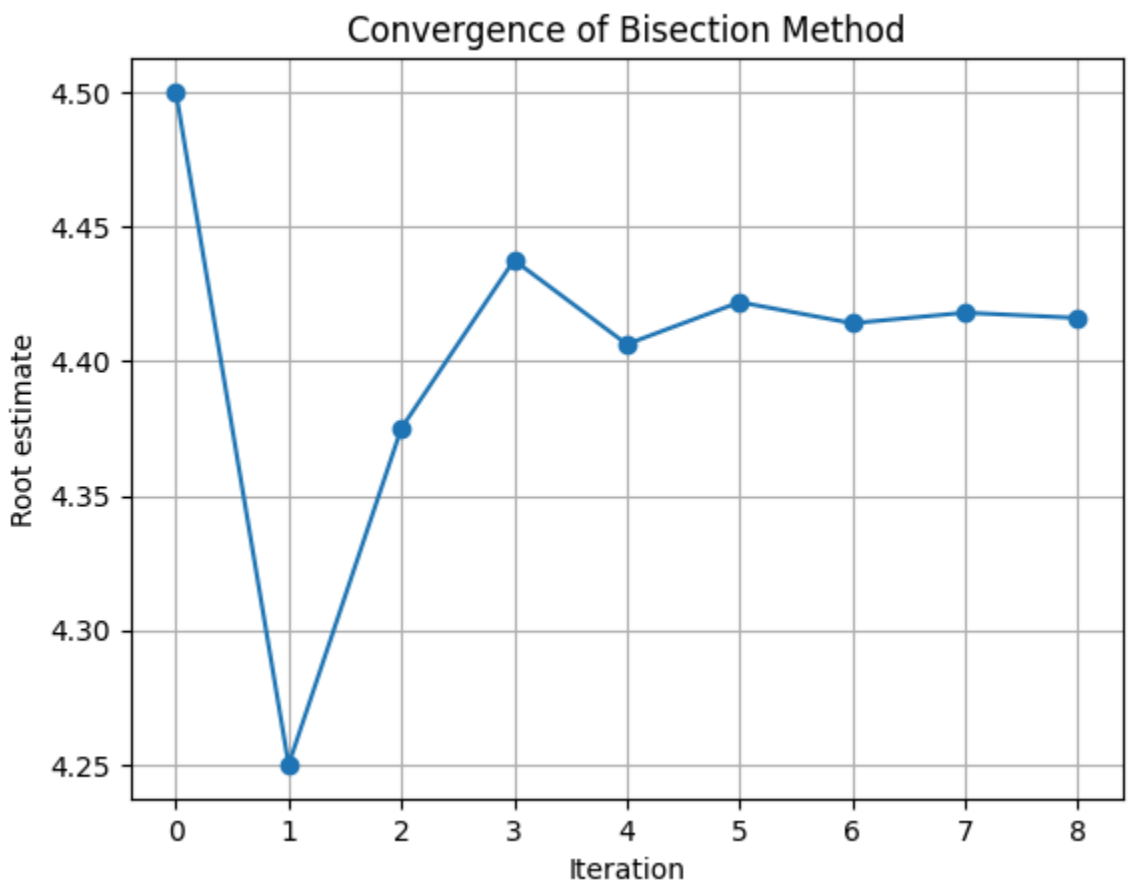


```
#we can say from the graph that , In the range [0, 7] f(x) = 0 has 3 roots
and it is clear that roots lie between the ranges [0, 4], [4, 5] and [5,
7].So we will find the 3 different roots using the above method for the 3
different range.


#finding the roots of f(x) = 0 using bisection method
eps = 1e-3
Nmax = 100  #Maximum number of iterations
a, b = 0, 4

# a, b = 4, 5

# a, b = 5, 7

#Define Bisection Method
def Bisection(Xminus, Xplus, Nmax, eps):
    roots = []

    for it in range(Nmax):
        x = (Xplus + Xminus) / 2.0
        roots.append(x)

        if f(Xplus) * f(x) > 0.0:
            Xplus = x
        else:
            Xminus = x

        if abs(f(x)) < eps:
            print("Converged in", it+1, "iterations")
            break

    return np.array(roots)
roots = Bisection(a, b, Nmax, eps)

plt.figure()
plt.plot(range(len(roots)), roots, marker='o')
plt.xlabel("Iteration")
plt.ylabel("Root estimate")
plt.title("Convergence of Bisection Method")
```

```
plt.grid(True)
plt.show()

print(roots)
print(f'root of the f(x) is {roots[-1]}')   #3.015625
print(f'value of f(x) at the root: {f(roots[-1])}')
#0.00022187845797638772

#output
Roots: [2.        3.        3.5       3.25      3.125     3.0625    3.03125
3.015625]
root of the f(x) is 3.015625
value of f(x) at the root: 0.00022187845797638772


Converged in 8 iterations
```

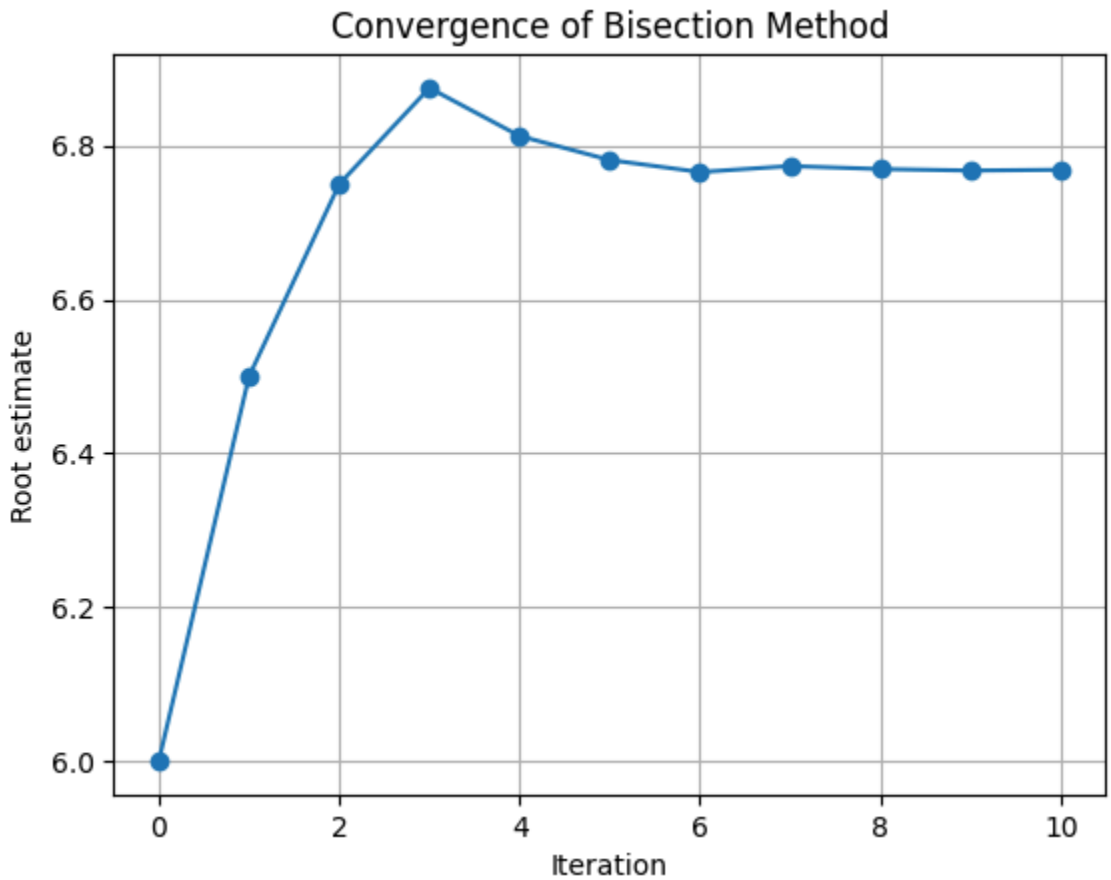


```
#for the range [4, 5] we will set a = 4 and b = 5
a, b = 4, 5

#Output:
Roots: [4.5        4.25       4.375      4.4375     4.40625    4.421875
 4.4140625  4.41796875 4.41601562]
root of the f(x) is 4.416015625
value of f(x) at the root: -0.00012283480382890843
is value of the fn at the root less then eps: True

Converged in 9 iterations
```




```
#for the range [5, 7] we will set a = 5 and b = 7
a, b = 5, 7

#Output:
Roots: [6.         6.5        6.75       6.875      6.8125     6.78125
```

```
 6.765625   6.7734375  6.76953125 6.76757812 6.76855469]
root of the f(x) is 6.7685546875
value of f(x) at the root: 0.00045067363184347897


Converged in 11 iterations
```




```
So the roots roots of the function are: 3.015625, 4.416015625,
6.7685546875
```

I really enjoyed doing this assignment:)